

Recognising Geometric Series, p-Series, and the nth-Term Test

Directions

- Three tools, three questions. *Is it geometric?* Look for a constant ratio r ; it converges when $|r| < 1$, to $\frac{\text{first term}}{1-r}$. *Is it a p-series?* $\sum \frac{1}{n^p}$ converges exactly when $p > 1$. *Do the terms go to zero?* If not, the series diverges.
- The nth-term test only ever proves *divergence*. Terms shrinking to zero proves nothing on its own.
- Where a table of partial sums is given, use it as a picture of what the symbols are doing — then answer with the algebra.

1. The table shows the first few partial sums of $\sum_{n=0}^{\infty} \left(\frac{1}{3}\right)^n$.

last index n	0	1	2	3
partial sum	1	1.3333	1.4444	1.4815

Find the exact sum of the whole series, and say what the table suggests about it.

Answer: _____

2. Find the exact sum of $\sum_{n=1}^{\infty} \frac{1}{n^2}$. (This is the most famous convergent p -series; you are expected to know its value.)

Answer: _____

3. Consider $\sum_{n=1}^{\infty} \frac{3n^2 + 1}{5n^2 + 2n}$.

- (a) Find $\lim_{n \rightarrow \infty} \frac{3n^2 + 1}{5n^2 + 2n}$.
- (b) State what the n th-term test lets you conclude about the series, and explain what it would *not* have told you had the limit been zero.

Answer: _____

4. A ball is dropped from a height of 2 metres. Each time it hits the floor it rebounds to $\frac{3}{5}$ of the height it fell from, and this continues forever. Find the total vertical distance the ball travels (down, then up and down again on every bounce).

Answer: _____ m

5. The table shows partial sums of $\sum_{n=1}^{\infty} \frac{1}{n^4}$.

last index n	1	2	3	4
partial sum	1	1.0625	1.0748	1.0788

- (a) Find the exact sum of the series.
- (b) Explain how the exponent alone tells you this series converges, and connect that to what the table is doing.

Answer: _____

6. Consider $\sum_{n=1}^{\infty} n \sin\left(\frac{1}{n}\right)$.

(a) Find $\lim_{n \rightarrow \infty} n \sin\left(\frac{1}{n}\right)$.

(b) State the conclusion the nth-term test gives, and explain why the terms fail to shrink even though $\sin\left(\frac{1}{n}\right)$ does.

Answer: _____

7. The repeating decimal $0.\overline{45} = 0.454545\dots$ can be written as the geometric series $\sum_{n=1}^{\infty} \frac{45}{100^n}$.

(a) Sum the series to write $0.\overline{45}$ as a fraction in lowest terms.

(b) Explain why the same argument shows that *every* repeating decimal is a rational number.

Answer: _____

8. Here are three series. Exactly one converges.

$$(i) \sum_{n=0}^{\infty} \left(\frac{2}{3}\right)^n \quad (ii) \sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} \quad (iii) \sum_{n=1}^{\infty} \frac{n}{n+4}$$

(a) Find the exact sum of series (i).

(b) Find $\lim_{n \rightarrow \infty} \frac{n}{n+4}$.

(c) Classify all three series, naming the deciding quantity in each case.

Answer: _____